Summary of paper 02: carryover representation and inference for biomarker-treatment interaction tests in aggregated N-of-1 trials under the correlation encoding of the interaction

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Problem and approach

Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Drug carryover attenuates the biomarker-by-treatment interaction. The analyst must decide how to represent residual exposure, how to obtain a correctly sized test, and whether to guard against a prognostic biomarker. We ran an exploratory, staged simulation study under the extended DGP, in which the interaction is a correlation between the biomarker and the drug response. Stage 1 compares three representations of residual exposure over 162 configurations. Stage 2 selects the half-life by AIC. Stage 3 adds CR2 cluster-robust inference and measures the cost of selection. Stage 4 adds the original analysis and an expectancy-term family. Stage 5 stresses the result with prognostic correlations, dropout and random slopes. The study ends in analysis G9: the exposure-weighted predictor with an AIC-selected half-life, refitted by REML and tested with CR2.

Simulation design

Estimand and analysis specifications

Table 1 defines the nine specifications, G1 to G9. G1 to G3 represent residual exposure, G4 selects the half-life, and G6 to G9 add CR2 inference.

Results

Stage 1: representing residual exposure

Table 2 gives the reference cell: Hybrid design, $N = 70$, biomarker-response correlation 0.45. At a half-life of 1 week, power is 0.806 for G1, 0.808 for G2 and 0.888 for G3.

Table 1: The nine G-family analysis specifications. G1-G3 are the Stage 1 representations of residual exposure; G4 replaces the assumed half-life by a data-driven choice (Stage 2); G6-G9 are CR2 cluster-robust variants of G1, G2, G4 and G3 (Stage 3). G5 collapses to one row per patient before fitting and has no CR2 variant. The external comparators E0 and the F-family are defined in Section
refsec-comparators.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc

Table 2: Stage 1: performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is mean(estimate $-\ \theta_{\text{true}}$) on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$; see the estimand-mismatch caveat in the text for G1/G2. Empirical SE is the standard deviation of the point estimates across replicates. Coverage is the proportion of nominal 95% Wald CIs containing θ_{true}. MSE is bias^2 + Empirical SE^2. Non-conv is the fraction of replicates where ‘nlme::lme()‘ failed to converge. MCSEs follow Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.994 (0.003)	-0.021 (0.031)	0.699	0.990 (0.004)	0.489 (0.029)	0.000
0.0	Lag-adjusted	0.994 (0.003)	-0.022 (0.031)	0.698	0.990 (0.004)	0.487 (0.029)	0.000
0.0	Exposure-weighted	0.994 (0.003)	-0.021 (0.031)	0.699	0.990 (0.004)	0.489 (0.029)	0.000
0.5	Unadjusted	0.946 (0.010)	+0.618 (0.034)	0.760	0.938 (0.011)	0.959 (0.055)	0.000
0.5	Lag-adjusted	0.944 (0.010)	+0.619 (0.034)	0.762	0.936 (0.011)	0.964 (0.055)	0.000
0.5	Exposure-weighted	0.962 (0.009)	+0.069 (0.039)	0.873	0.970 (0.008)	0.766 (0.052)	0.000
1.0	Unadjusted	0.806 (0.018)	+1.143 (0.036)	0.799	0.790 (0.018)	1.943 (0.092)	0.000
1.0	Lag-adjusted	0.808 (0.018)	+1.143 (0.036)	0.808	0.786 (0.018)	1.959 (0.092)	0.000
1.0	Exposure-weighted	0.888 (0.014)	-0.084 (0.047)	1.062	0.964 (0.008)	1.135 (0.074)	0.000

Figure 1 shows power against the half-life. The G1 and G2 curves coincide, and G3 falls less as the half-life grows.

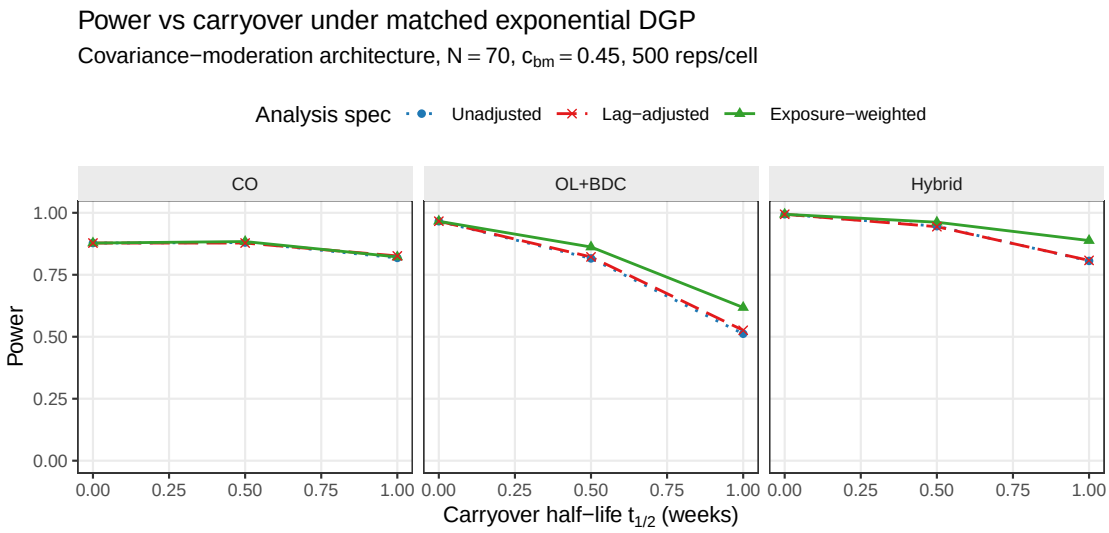


Figure 1: Stage 1: power versus carryover half-life under the exponential DGP, by analysis specification and trial design ($c_{bm} = 0.45$, $n_{sim} = 500$).

Stage 2: removing the analyst's half-life

Table 3 adds G4, which chooses the half-life from the data. For Hybrid, $N = 70$, $c_{bm} = 0.45$ and $t_{1/2} = 1$, power is 0.790 for G1, 0.886 for G3 and 0.876 for G4.

Stage 3: a correctly sized test

Table 4 shows the model-based test rejecting at 0.033 to 0.046 under the null. CR2 moves the rejection rate to 0.046 to 0.059, and G4 is highest.

Table 5 pools the four null cells. G9 rejects at 0.0555 against 0.0493 for G8, a paired difference of 0.0059 (MCSE 0.0013).

Table 6 shows that AIC selection does not recover the half-life. At a true half-life of 0.5 week and $N = 70$, the smallest candidate is chosen in 0.59 of replicates and 0.5 week in 0.33.

Selection on the null model and fixed half-lives

Table 7 compares selection variants. At $t_{1/2} = 1.0$, $N = 35$ and $c_{PB} = 0$, G9 rejects at 0.0596, G9-null at 0.0506 and G8 at 0.0508.

Table 8 gives power. At a true half-life of 2 weeks and $N = 70$ with $c_{bm} = 0.45$, power is 0.679 for G8, 0.515 for G9, 0.667 for G9-null and 0.661 for G8 fixed at 1 week. The fixed 1-week analysis loses at most 0.022 against G8 in every row.

Table 3: Stage 2: power of the binary indicator (G1), the exposure-weighted predictor at the true half-life (G3) and the exposure-weighted predictor at the AIC-selected half-life (G4), model-based inference, on the 30-cell corrected grid (exponential DGP, $n_{sim} = 500$, same simulated trial for all three each replicate). Two-MCSE half-width is about 0.04 at $p = 0.5$.

Design	N	c_{bm}	$t_{1/2}$	G1	G3	G4
CO	35	0.45	0.0	0.582	0.582	0.582
CO	35	0.45	0.5	0.576	0.572	0.580
CO	35	0.30	1.0	0.280	0.280	0.294
CO	35	0.45	1.0	0.514	0.524	0.532
CO	70	0.45	0.0	0.880	0.880	0.878
CO	70	0.45	0.5	0.874	0.874	0.874
CO	70	0.30	1.0	0.508	0.518	0.522
CO	70	0.45	1.0	0.832	0.852	0.844
Hybrid	35	0.45	0.0	0.826	0.826	0.816
Hybrid	35	0.45	0.5	0.656	0.678	0.688
Hybrid	35	0.30	1.0	0.212	0.250	0.276
Hybrid	35	0.45	1.0	0.450	0.524	0.534
Hybrid	70	0.45	0.0	0.992	0.992	0.990
Hybrid	70	0.45	0.5	0.960	0.960	0.962
Hybrid	70	0.30	1.0	0.448	0.518	0.534
Hybrid	70	0.45	1.0	0.790	0.886	0.876
OL+BDC	35	0.45	0.0	0.694	0.694	0.700
OL+BDC	35	0.45	0.5	0.474	0.506	0.523
OL+BDC	35	0.30	1.0	0.134	0.164	0.182
OL+BDC	35	0.45	1.0	0.248	0.330	0.344
OL+BDC	70	0.45	0.0	0.958	0.958	0.960
OL+BDC	70	0.45	0.5	0.820	0.842	0.842
OL+BDC	70	0.30	1.0	0.276	0.338	0.346
OL+BDC	70	0.45	1.0	0.544	0.638	0.639

Table 4: Stage 3: conservatism of the model-based interaction test and its cluster-robust correction at four Hybrid null cells ($c_{bm} = 0$, exponential DGP, corrected simulation code, 2,000 replicates each). κ is the calibration ratio of Section 2.4; $\kappa > 1$ is a conservative test. The CR2 columns are G6, G8 and G9 for G1, G3 and G4 respectively. Two-MCSE band around 0.05: $[0.040, 0.060]$.

$t_{1/2}$	N	Specification	κ model	Type I model	κ CR2	Type I CR2
0.5	35	G1	1.05	0.041	0.97	0.049
0.5	35	G3	1.05	0.035	0.97	0.050
0.5	35	G4	1.00	0.043	0.93	0.056
0.5	70	G1	1.05	0.042	0.98	0.050
0.5	70	G3	1.06	0.037	0.99	0.050
0.5	70	G4	1.05	0.040	0.98	0.052
1.0	35	G1	1.05	0.040	0.97	0.047
1.0	35	G3	1.06	0.033	0.98	0.052
1.0	35	G4	0.99	0.046	0.91	0.059
1.0	70	G1	1.05	0.041	0.98	0.049
1.0	70	G3	1.07	0.035	0.99	0.046
1.0	70	G4	1.03	0.042	0.96	0.055

Table 5: Stage 3: null rejection of G6, G8 and G9 pooled over the four 2,000-replicate Hybrid null cells of Table
reftab:tab-type1-cr2; the paired difference between G9 and G8 on the same trials (the discordant counts are 77 replicates rejected by G9 only and 30 by G8 only); and G9 conditional on the half-life it selected. From '34-referee-reanalysis.R'.

Quantity	Method	n	Rate	MCSE
Pooled null rejection	G6	7986	0.0486	0.0024
Pooled null rejection	G8	7988	0.0493	0.0024
Pooled null rejection	G9	7987	0.0555	0.0026
Paired difference, G9 minus G8	G9-G8	7980	0.0059	0.0013
G9 given selected $\hat{t}^a_{1/2} = 0.25$	G9	3116	0.0507	0.0039
G9 given selected $\hat{t}^a_{1/2} = 0.5$	G9	2232	0.0587	0.0050
G9 given selected $\hat{t}^a_{1/2} = 1$	G9	1613	0.0583	0.0058
G9 given selected $\hat{t}^a_{1/2} = 2$	G9	1026	0.0585	0.0073

Table 6: Stage 3: share of replicates in which each candidate half-life (weeks) was selected by AIC, Hybrid design, pooled over every stored cell at the given true half-life and N (null and alternative cells together). From '34-referee-reanalysis.R'.

			Selected half-life			
True $t_{1/2}$	N	n	0.25	0.5	1	2
0.0	35	500	0.97	0.02	0.00	0.00
0.0	70	500	0.99	0.01	0.00	0.00
0.2	35	1000	0.95	0.05	0.00	0.00
0.2	70	992	0.98	0.02	0.00	0.00
0.5	35	34957	0.56	0.29	0.11	0.04
0.5	70	47337	0.59	0.33	0.06	0.01
1.0	35	3500	0.20	0.24	0.29	0.28
1.0	70	3493	0.12	0.26	0.40	0.21
2.0	35	1000	0.16	0.07	0.14	0.63
2.0	70	1992	0.06	0.06	0.15	0.73
4.0	35	1000	0.26	0.06	0.07	0.60
4.0	70	991	0.18	0.03	0.07	0.72

Table 7: Stage 3, referee-variant runs: null rejection rate ($c_{bm} = 0$) of G8, G9, G9-null, G8 at two fixed analysis half-lives and F1 in eight Hybrid null cells, 5,000 replicates each (MCSE about 0.003), independent seeds per cell, with the paired per-replicate difference against G8 (paired SE 0.0015 to 0.0021) and the calibration ratio. c_{PB}: a prognostic correlation with the placebo-response component.

Cell	Method	Rejection	Paired diff. vs G8	κ
$t_{1/2} = 0.5$, $N = 35$, $c_{PB} = 0.0$	G8	0.0482	–	0.97
	G9	0.0522	+0.0040	0.93
	G9-null	0.0470	-0.0012	0.97
	G8 at 0.25 wk	0.0460	-0.0022	0.98
	G8 at 1.00 wk	0.0462	-0.0020	0.96
	F1	0.0466	-0.0016	0.98
$t_{1/2} = 0.5$, $N = 35$, $c_{PB} = 0.3$	G8	0.0524	–	0.96
	G9	0.0584	+0.0060	0.91
	G9-null	0.0532	+0.0008	0.95
	G8 at 0.25 wk	0.0544	+0.0020	0.96
	G8 at 1.00 wk	0.0510	-0.0014	0.96
	F1	0.0526	+0.0002	0.96
$t_{1/2} = 0.5$, $N = 70$, $c_{PB} = 0.0$	G8	0.0514	–	0.99
	G9	0.0538	+0.0024	0.97
	G9-null	0.0510	-0.0004	0.98
	G8 at 0.25 wk	0.0521	+0.0008	0.98
	G8 at 1.00 wk	0.0492	-0.0022	0.99
	F1	0.0487	-0.0026	0.99
$t_{1/2} = 0.5$, $N = 70$, $c_{PB} = 0.3$	G8	0.0503	–	0.99
	G9	0.0544	+0.0038	0.98
	G9-null	0.0508	+0.0002	1.00
	G8 at 0.25 wk	0.0494	-0.0010	0.99
	G8 at 1.00 wk	0.0508	+0.0002	0.99
	F1	0.0492	-0.0016	0.99
$t_{1/2} = 1.0$, $N = 35$, $c_{PB} = 0.0$	G8	0.0508	–	0.96
	G9	0.0596	+0.0088	0.89
	G9-null	0.0506	-0.0002	0.94
	G8 at 0.25 wk	0.0504	-0.0004	0.96
	G8 at 1.00 wk	0.0508	+0.0000	0.96
	F1	0.0512	+0.0004	0.97
$t_{1/2} = 1.0$, $N = 35$, $c_{PB} = 0.3$	G8	0.0432	–	0.98
	G9	0.0536	+0.0104	0.91
	G9-null	0.0422	-0.0010	0.96
	G8 at 0.25 wk	0.0416	-0.0016	0.99
	G8 at 1.00 wk	0.0432	+0.0000	0.98
	F1	0.0380	-0.0052	0.99
$t_{1/2} = 1.0$, $N = 70$, $c_{PB} = 0.0$	G8	0.0461	–	1.00
	G9	0.0505	+0.0044	0.95
	G9-null	0.0447	-0.0012	0.99
	G8 at 0.25 wk	0.0427	-0.0036	1.00
	G8 at 1.00 wk	0.0461	+0.0000	1.00
	F1	0.0428	-0.0034	1.00
$t_{1/2} = 1.0$, $N = 70$, $c_{PB} = 0.3$	G8	0.0554	–	0.95
	G9	0.0634	+0.0080	0.90
	G9-null	0.0568	+0.0012	0.94
	G8 at 0.25 wk	0.0567	+0.0010	0.95
	G8 at 1.00 wk	0.0554	+0.0000	0.95
	F1	0.0546	-0.0010	0.95

Table 8: Stage 3, referee-variant runs: power at eight Hybrid alternative cells, 1,000 replicates each (MCSE at most 0.016), independent seeds per cell, for the same analyses as the null table. The true half-life is 0.5 or 2 weeks; the fixed analysis half-lives are 0.25 and 1 week in every cell.

$t_{1/2}$	N	c_{bm}	G8	G9	G9-null	G8 at 0.25 wk	G8 at 1 wk	F1
0.5	35	0.25	0.291	0.276	0.273	0.293	0.277	0.267
0.5	70	0.25	0.538	0.517	0.517	0.538	0.526	0.514
0.5	35	0.45	0.743	0.735	0.723	0.753	0.737	0.715
0.5	70	0.45	0.966	0.958	0.959	0.964	0.960	0.951
2.0	35	0.25	0.124	0.097	0.117	0.139	0.108	0.088
2.0	70	0.25	0.248	0.187	0.256	0.271	0.239	0.180
2.0	35	0.45	0.384	0.307	0.370	0.410	0.362	0.288
2.0	70	0.45	0.679	0.515	0.667	0.690	0.661	0.483

Table 9: Stage 3: Type I error at the six null cells of the corrected grid ($c_{bm} = 0$, $t_{1/2} = 1$ week, exponential DGP, $n_{sim} = 500$). Each model-based specification is followed by its CR2 variant (G1/G6, G3/G8, G4/G9). E0 is the original analysis. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Design	N	G1	G6	G3	G8	G4	G9	E0
CO	35	0.030	0.046	0.034	0.052	0.042	0.058	0.150
CO	70	0.046	0.054	0.046	0.050	0.050	0.056	0.128
Hybrid	35	0.042	0.066	0.030	0.044	0.042	0.060	0.070
Hybrid	70	0.032	0.046	0.034	0.054	0.038	0.058	0.060
OL+BDC	35	0.048	0.074	0.048	0.052	0.044	0.056	0.144
OL+BDC	70	0.030	0.046	0.030	0.048	0.032	0.050	0.106

Table 9 returns to the six corrected-grid null cells at $t_{1/2} = 1$ week. G1, G3 and G4 reject at 0.030 to 0.050, the CR2 variants at 0.044 to 0.074, and E0 at 0.060 to 0.150.

Figure 2 shows power for the nine specifications at $t_{1/2} = 1$ week. In the Hybrid design at $N = 35$, G8 and G9 reach 0.54 at $c_{bm} = 0.45$.

Figure 3 varies the half-life at $c_{bm} = 0.45$. Read the loss in power from left to right within each design.

Stage 4: external comparators on the same trials

Table 10 compares G9 with E0 and the expectancy-term family under the null. Under OL+BDC, E0 rejects at 0.100 and G9 at 0.060, and G9 stays at 0.048 to 0.052 on the prognostic rows.

Table 11 gives power. At $t_{1/2} = 2$ weeks and $c_{bm} = 0.25$, G9 reaches 0.281 against 0.192 for F1 and 0.228 for F1A0B.

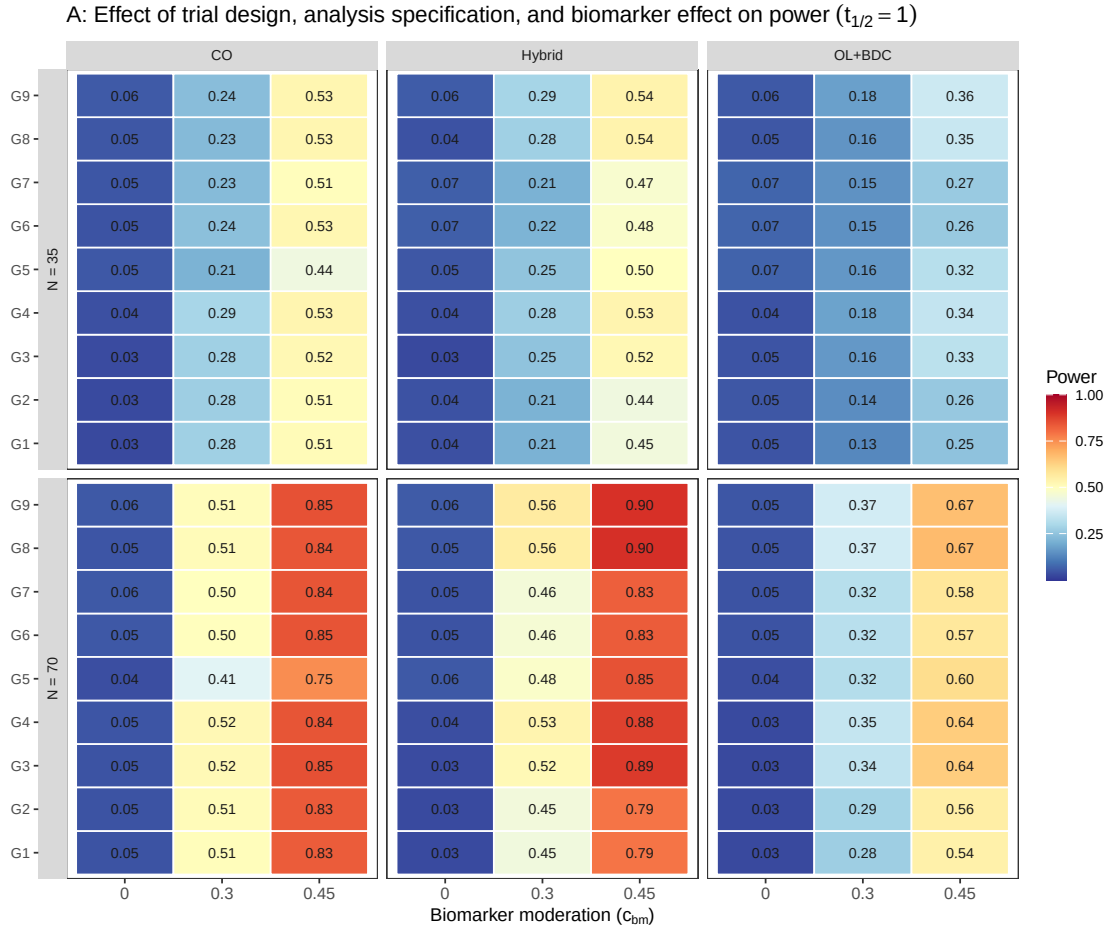


Figure 2: Stage 3: power by trial design (columns), N (row blocks) and analysis specification (rows within block) across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, corrected simulation code ($n_{sim} = 500$). The $c_{bm} = 0$ column is Type I error.

Table 10: Stage 4: Type I error ($c_{bm} = 0$) at eight null scenarios on Hybrid ($N = 70$, $t_{1/2} = 0.5$ week unless stated; OL+BDC row is that design), $n_{sim} = 500$, same trial for every specification each replicate. The prognostic rows set one pathway parameter to 0.3; the prog-TV-grow row is a constant shift absorbed by the biomarker main effect and is identical to the Hybrid null row by construction. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Hybrid	0.034	0.032	0.034	0.052	0.046	0.048	0.060	0.058	0.046
OL+BDC	0.030	0.036	0.036	0.051	0.048	0.060	0.100	0.050	0.038
prog-BL 0.3	0.034	0.032	0.034	0.054	0.046	0.048	0.062	0.059	0.047
prog-TV-const 0.3	0.034	0.034	0.036	0.052	0.046	0.048	0.066	0.061	0.048
prog-TV-grow 0.3	0.034	0.032	0.034	0.052	0.046	0.048	0.060	0.058	0.046
prog-PB 0.3	0.034	0.030	0.032	0.056	0.048	0.052	0.074	0.058	0.046
$t_{1/2} = 2$ wk	0.034	0.038	0.042	0.052	0.054	0.062	0.066	0.059	0.047
Weibull $k = 2$ data	0.032	0.032	0.032	0.052	0.046	0.052	0.060	0.058	0.058

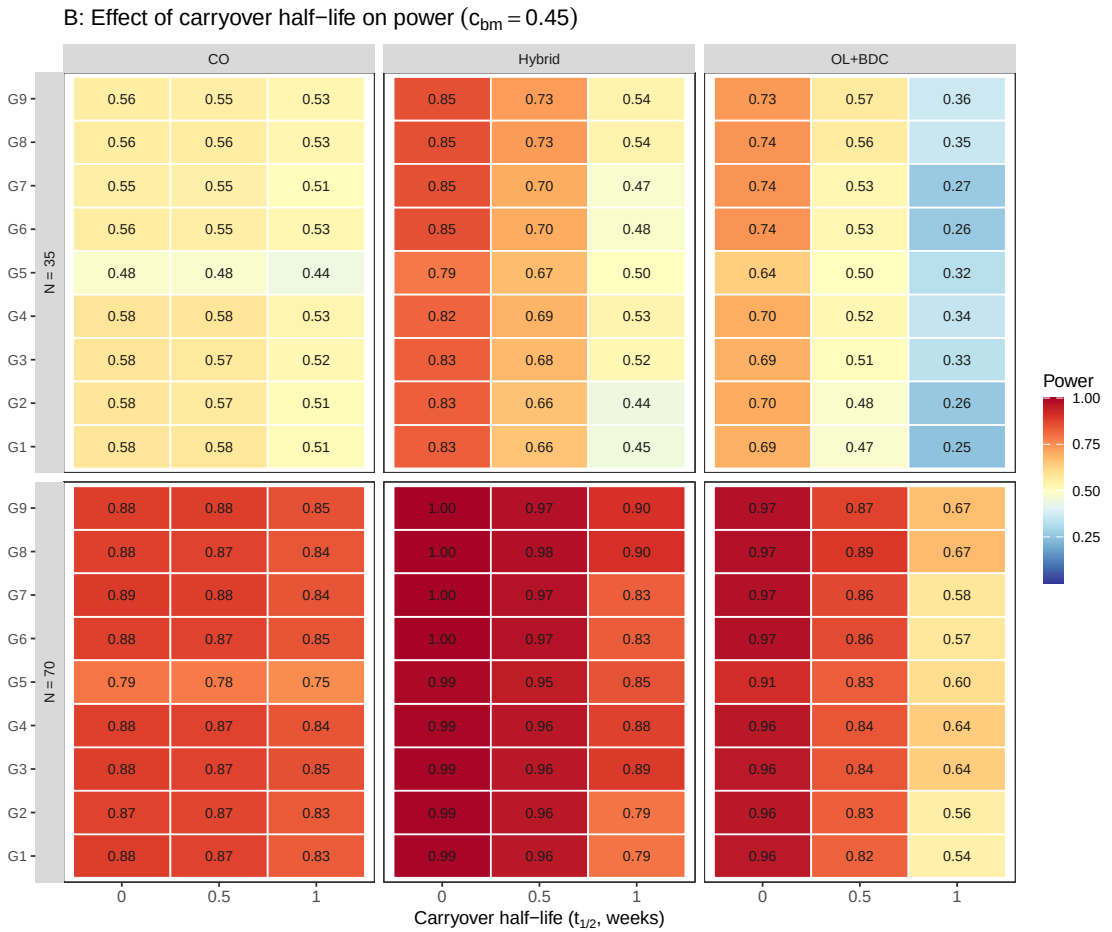


Figure 3: Stage 3: power across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, corrected simulation code ($n_{sim} = 500$).

Table 11: Stage 4: power at five alternative scenarios (same specifications and n_{sim} as the null table). G6 and G8 are the equal-size comparators for G9; E0 is liberal and its power is not comparable.

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Hybrid, $c_{bm} = 0.45$	0.934	0.950	0.950	0.962	0.972	0.968	0.932	0.952	0.950
Hybrid, $c_{bm} = 0.25$	0.510	0.542	0.544	0.564	0.574	0.584	0.546	0.538	0.500
OL+BDC, $c_{bm} = 0.45$	0.842	0.842	0.825	0.865	0.871	0.863	0.866	0.848	0.812
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.180	0.242	0.265	0.196	0.257	0.281	0.284	0.192	0.228
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.630	0.610	0.631	0.653	0.655	0.661	0.612	0.652	0.614

Table 12: Stage 4, scenario grids: Type I error ($c_{bm} = 0$) on Hybrid at $N \in \{35, 70\}$ under a Weibull decay analyzed as exponential, a true half-life below, at the boundary of and above G4's candidate grid, and two autocorrelations ($t_{1/2} = 0.5$ week and $\rho = 0.7$ unless stated; $n_{sim} = 500$; band $[0.031, 0.069]$).

Scenario	N	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Weibull $k = 0.5$ data	35	0.038	0.040	0.048	0.040	0.042	0.052	0.054	0.044	0.044
Weibull $k = 0.5$ data	70	0.026	0.028	0.028	0.034	0.040	0.048	0.060	0.042	0.050
Weibull $k = 2$ data	35	0.038	0.032	0.040	0.040	0.042	0.038	0.054	0.044	0.032
Weibull $k = 2$ data	70	0.026	0.028	0.030	0.036	0.042	0.038	0.062	0.043	0.049
$t_{1/2} = 0.2$ wk	35	0.038	0.040	0.040	0.040	0.038	0.038	0.054	0.044	0.030
$t_{1/2} = 0.2$ wk	70	0.026	0.026	0.028	0.036	0.032	0.034	0.060	0.044	0.047
$t_{1/2} = 2$ wk	35	0.036	0.022	0.034	0.042	0.050	0.056	0.048	0.046	0.028
$t_{1/2} = 2$ wk	70	0.026	0.030	0.034	0.034	0.046	0.058	0.056	0.042	0.056
$t_{1/2} = 4$ wk	35	0.036	0.022	0.044	0.042	0.044	0.064	0.050	0.046	0.030
$t_{1/2} = 4$ wk	70	0.026	0.026	0.038	0.036	0.042	0.067	0.060	0.042	0.046
$\rho = 0.5$	35	0.032	0.036	0.040	0.044	0.046	0.060	0.056	0.048	0.052
$\rho = 0.5$	70	0.024	0.030	0.028	0.030	0.044	0.046	0.062	0.044	0.034
$\rho = 0.9$	35	0.016	0.018	0.016	0.038	0.036	0.034	0.040	0.034	0.030
$\rho = 0.9$	70	0.018	0.018	0.022	0.044	0.050	0.048	0.034	0.044	0.050

Scenario grids

Table 12 crosses the scenarios with two sample sizes. The G9 Type I error is 0.034 to 0.067 in all fourteen cells.

Table 13 gives power. At $\rho = 0.9$ and $N = 35$, G9 reaches 0.808 against 0.684 for G1 and 0.608 for E0.

Dropout

Table 14 imposes monotone dropout. The G9 Type I error is 0.026 to 0.062, and at rate 0.3 and $N = 35$ its power is 0.617 (MAR) and 0.649 (MCAR).

Table 15 gives interval coverage. G8 covers at 0.950 to 0.959 on average, and G9 at 0.934 to 0.947.

Table 16 is the heavy-missingness case. The conventional κ of G9 is 0.46 and 0.49, but the estimate SD of 12.4 is more than twice the MAD-based SD, and the robust κ is 0.78 and 0.86.

Stage 5: prognostic confounds, feasibility, and random effects

Prognostic confounds at two strengths

Table 17 gives size at strength 0.3. The G9 Type I error is 0.044 to 0.050 on the four pathways.

Table 18 gives power at two strengths. At strength 0.3 and $N = 35$ the G9 power is 0.778 to 0.798, against a no-confound reference of 0.759, and the caption marks the strength-0.6 cells as a data-generating artifact.

Table 13: Stage 4, scenario grids: power at one alternative per scenario (same cells and n_{sim} as the null table; $c_{bm} = 0.30$ for the autocorrelation rows because 0.45 is infeasible at $\rho = 0.9$).

Scenario	N	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	35	0.538	0.552	0.572	0.548	0.586	0.588	0.586	0.541	0.523
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	70	0.878	0.894	0.889	0.900	0.915	0.913	0.876	0.891	0.891
Weibull $k = 2$ data, $c_{bm} = 0.45$	35	0.830	0.814	0.828	0.850	0.848	0.850	0.794	0.826	0.809
Weibull $k = 2$ data, $c_{bm} = 0.45$	70	0.994	0.990	0.994	0.996	0.994	0.994	0.986	0.992	0.992
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	35	0.348	0.344	0.342	0.340	0.342	0.338	0.336	0.328	0.292
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	70	0.632	0.626	0.628	0.659	0.659	0.657	0.614	0.645	0.659
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	35	0.108	0.124	0.150	0.112	0.146	0.168	0.160	0.114	0.134
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	70	0.176	0.220	0.235	0.185	0.233	0.254	0.246	0.177	0.218
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	35	0.058	0.066	0.080	0.064	0.080	0.102	0.098	0.066	0.084
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	70	0.084	0.104	0.137	0.092	0.134	0.139	0.140	0.087	0.105
$\rho = 0.5$, $c_{bm} = 0.30$	35	0.280	0.300	0.310	0.284	0.288	0.306	0.358	0.264	0.248
$\rho = 0.5$, $c_{bm} = 0.30$	70	0.542	0.546	0.552	0.595	0.606	0.596	0.610	0.522	0.456
$\rho = 0.9$, $c_{bm} = 0.30$	35	0.684	0.734	0.717	0.785	0.808	0.808	0.608	0.794	0.772
$\rho = 0.9$, $c_{bm} = 0.30$	70	0.974	0.980	0.980	0.994	0.990	0.994	0.910	0.994	0.994

Table 14: Stage 4: Type I error and power under monotone dropout (Hybrid, $t_{1/2} = 0.5$ week; the rate is the fraction of patients with an incomplete series; $n_{sim} = 500$; band $[0.031, 0.069]$).

	Type I error				Power, $c_{bm} = 0.45$			
Dropout (rate, N)	G1	G8	G9	F1	G1	G8	G9	F1
MAR 0.1, $N = 35$	0.028	0.042	0.050	0.046	0.672	0.719	0.717	0.662
MAR 0.2, $N = 35$	0.034	0.046	0.052	0.048	0.638	0.687	0.687	0.607
MAR 0.3, $N = 35$	0.030	0.044	0.054	0.042	0.596	0.630	0.617	0.571
MCAR 0.1, $N = 35$	0.030	0.048	0.062	0.046	0.674	0.716	0.718	0.666
MCAR 0.2, $N = 35$	0.024	0.040	0.054	0.044	0.644	0.686	0.681	0.632
MCAR 0.3, $N = 35$	0.024	0.036	0.042	0.038	0.608	0.646	0.649	0.584
MAR 0.1, $N = 70$	0.026	0.040	0.040	0.042	0.956	0.980	0.976	0.958
MAR 0.2, $N = 70$	0.022	0.038	0.042	0.040	0.936	0.972	0.976	0.932
MAR 0.3, $N = 70$	0.038	0.040	0.042	0.036	0.910	0.940	0.948	0.920
MCAR 0.1, $N = 70$	0.022	0.036	0.040	0.042	0.958	0.978	0.978	0.952
MCAR 0.2, $N = 70$	0.020	0.032	0.026	0.026	0.950	0.976	0.970	0.956
MCAR 0.3, $N = 70$	0.024	0.032	0.040	0.030	0.926	0.956	0.952	0.924

Table 15: Coverage of the nominal 95 percent CR2 interval for G8 and G9 over the alternative cells of the supplementary study, on the calibrated scale ($\theta = -c_{bm}\sigma_{BR}$). The Satterthwaite degrees of freedom are not stored per replicate and were recovered from the stored t and p where $|t| \geq 0.5$; the remaining replicates are classified as covering, not covering, or ambiguous, which gives the lower and upper bounds. From '34-referee-reanalysis.R'.

Method	N	Cells	Mean coverage (lower, upper)	Minimum (lower)
G8	35	21	0.950, 0.959	0.938
G8	70	28	0.950, 0.954	0.915
G9	35	21	0.938, 0.947	0.890
G9	70	28	0.934, 0.940	0.873

Table 16: Heavy-missingness stress case (per-occasion MCAR 0.3, $N = 35$, 500 replicates): dispersion of the interaction estimates on the calibrated scale and the calibration ratio on the conventional scale (κ, mean SE over SD) and a robust one (median SE over 1.4826 times the median absolute deviation). From '34-referee-reanalysis.R'.

Cell	Method	SD of estimates	MAD-based SD	Max $\lvert\hat{\beta}\rvert$	κ	Robust κ	Rejection
$c_{bm} = 0.45$	G6	6.1	3.1	89	0.58	0.94	0.075
$c_{bm} = 0.45$	G8	7.5	3.9	110	0.57	0.91	0.085
$c_{bm} = 0.45$	G9	12.4	5.2	137	0.46	0.78	0.083
$c_{bm} = 0.45$	F1	6.1	3.2	86	0.59	0.93	0.075
null	G6	6.1	3.3	82	0.61	0.90	0.020
null	G8	7.6	4.2	102	0.60	0.88	0.024
null	G9	12.4	5.1	131	0.49	0.86	0.022
null	F1	6.2	3.5	78	0.62	0.92	0.024

Table 17: Stage 5: Type I error of G9 and F1A0B under each prognostic pathway at confound strength 0.3, with no true interaction ($c_{bm} = 0$), Hybrid, $t_{1/2} = 0.5$ week, $n_{sim} = 500$ (band $[0.031, 0.069]$). The strength-0.6 cells on prog-BL, prog-PB and prog-TV-const were generated under a repaired covariance matrix, that is, a different DGP, and are not evidence about size (see text). The constant-shift row is identical to the no-confound null by construction.

	G9		F1A0B	
Pathway	$N = 35$	$N = 70$	$N = 35$	$N = 70$
prog-BL	0.048	0.044	0.044	0.034
prog-TV-const	0.050	0.048	0.046	0.034
prog-TV-shift (constant)	0.046	0.046	0.046	0.034
prog-PB	0.046	0.044	0.046	0.034

Table 18: Stage 5: power of G9 and F1A0B with a true interaction ($c_{bm} = 0.45$) and a co-occurring prognostic confound, same layout as the size table. The no-confound reference for G9 is 0.759 at $N = 35$ and 0.969 at $N = 70$ (1,000 replicates, a different job from these cells, so the comparison is unpaired). Strength-0.6 cells on prog-BL, prog-PB and prog-TV-const are a data-generating artifact, not a confound effect (see text).

	G9, $N = 35$		G9, $N = 70$		F1A0B, $N = 35$		F1A0B, $N = 70$	
Pathway	0.3	0.6	0.3	0.6	0.3	0.6	0.3	0.6
prog-BL	0.782	0.758	0.976	0.960	0.738	0.702	0.950	0.920
prog-TV-const	0.782	0.730	0.968	0.950	0.502	0.482	0.772	0.702
prog-TV-shift (constant)	0.778	0.756	0.970	0.966	0.778	0.756	0.970	0.966
prog-PB	0.798	0.741	0.968	0.940	0.597	0.562	0.853	0.794

Table 19: Stage 5: positive definiteness of the raw joint covariance matrix on the four Hybrid paths, and the population on-drug minus off-drug slope contrast of the outcome on the biomarker before and after the silent repair, for the prognostic cells at $c_{bm} = 0.45$ and the strength-0.6 null cells (computed from the matrix, no simulation). The contrast is the population value of the interaction term.

Pathway	Strength	c_{bm}	Paths repaired	Min. eigenvalue	Contrast, raw	Contrast, repaired
none	0.0	0.45	0	0.180	-3.26	-3.26
prog-BL	0.3	0.45	0	0.145	-3.26	-3.26
prog-BL	0.6	0.00	0	0.184	0.00	0.00
prog-BL	0.6	0.45	4	-0.110	-3.26	-3.00
prog-PB	0.3	0.45	0	0.010	-3.40	-3.40
prog-PB	0.6	0.00	4	-0.240	-0.27	-0.23
prog-PB	0.6	0.45	4	-0.642	-3.53	-2.38
prog-TV-const	0.3	0.45	0	0.011	-3.26	-3.26
prog-TV-const	0.6	0.00	4	-0.354	0.00	0.00
prog-TV-const	0.6	0.45	4	-0.853	-3.26	-1.81

Table 20: Stage 5: exact feasibility bounds (the ‘Ceiling‘ column) for each biomarker correlation (the ‘Coupling‘ column), alone and given a fixed biomarker-response correlation c_{bm}, Hybrid design (minimum over paths), $t_{1/2} = 0.5$ week, $\rho = 0.7$, $c_1 = 0.2$. Each bound is verified by the smallest eigenvalue of the full correlation matrix being positive at 0.999 times and negative at 1.001 times the stated value.

Coupling	Given	Ceiling
c_{bm}	alone	0.518
c_{pb}	alone	0.513
c_{tv}	alone	0.513
c_{bl}	alone	1.000
c_{pb}	$c_{bm} = 0.3$	0.451
c_{tv}	$c_{bm} = 0.3$	0.451
c_{bl}	$c_{bm} = 0.3$	0.815
c_{pb}	$c_{bm} = 0.45$	0.306
c_{tv}	$c_{bm} = 0.45$	0.306
c_{bl}	$c_{bm} = 0.45$	0.496

The strength-0.6 loss is a data-generating artifact

Table 19 shows that every strength-0.6 cell with the interaction had a matrix that was not positive definite on all four paths. The repair moved the contrast from -3.53 to -2.38 for prog-PB and from -3.26 to -1.81 for prog-TV-const.

The feasibility bound for a prognostic correlation

Table 20 gives the closed-form bounds. The bound for c_{pb} falls from 0.513 alone to 0.451 given $c_{bm} = 0.3$, so strength 0.6 is infeasible.

Table 21: Stage 5: rejection rate on the feasible prognostic-confound ladder (Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, $n_{sim} = 500$). Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$); the others are power. Every cell passed the pre-flight positive-definiteness check.

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
prog-PB 0.40, $c_{bm} = 0.00$	0.030	0.028	0.028	0.040	0.042	0.044	0.064	0.043	0.059
prog-PB 0.40, $c_{bm} = 0.30$	0.706	0.714	0.721	0.737	0.752	0.757	0.750	0.680	0.713
prog-PB 0.45, $c_{bm} = 0.00$	0.028	0.030	0.030	0.040	0.040	0.044	0.062	0.040	0.048
prog-PB 0.45, $c_{bm} = 0.30$	0.694	0.730	0.729	0.751	0.776	0.780	0.756	0.694	0.712
prog-TV-const 0.40, $c_{bm} = 0.00$	0.024	0.028	0.022	0.040	0.038	0.046	0.060	0.046	0.046
prog-TV-const 0.40, $c_{bm} = 0.30$	0.678	0.680	0.687	0.692	0.734	0.729	0.694	0.671	0.695
prog-TV-const 0.45, $c_{bm} = 0.00$	0.028	0.028	0.026	0.038	0.038	0.044	0.058	0.046	0.045
prog-TV-const 0.45, $c_{bm} = 0.30$	0.662	0.680	0.684	0.704	0.720	0.721	0.706	0.679	0.685
prog-BL 0.45, $c_{bm} = 0.00$	0.026	0.028	0.026	0.032	0.042	0.046	0.058	0.047	0.059
prog-BL 0.45, $c_{bm} = 0.45$	0.958	0.970	0.970	0.976	0.976	0.978	0.958	0.968	0.972
prog-BL 0.60, $c_{bm} = 0.00$	0.030	0.032	0.030	0.030	0.040	0.043	0.058	0.045	0.059
prog-BL 0.60, $c_{bm} = 0.30$	0.656	0.660	0.665	0.681	0.708	0.694	0.684	0.648	0.675
prog-BL 0.80, $c_{bm} = 0.00$	0.026	0.030	0.028	0.034	0.042	0.042	0.060	0.046	0.050
prog-BL 0.80, $c_{bm} = 0.30$	0.638	0.666	0.664	0.679	0.688	0.682	0.692	0.650	0.660

Table 22: Stage 5: Type I error under prog-PB with no true interaction at larger samples (Hybrid, $t_{1/2} = 0.5$ week; $n_{sim} = 1{,}000$ at $N = 140$ and 280, band $[0.036, 0.064]$; the $N = 70$ row is from the ladder, $n_{sim} = 500$).

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
$c_{pb} = 0.30$, $N = 140$	0.044	0.043	0.043	0.058	0.055	0.055	0.068	0.057	0.049
$c_{pb} = 0.30$, $N = 280$	0.053	0.045	0.052	0.065	0.063	0.063	0.071	0.067	0.060
$c_{pb} = 0.45$, $N = 70$	0.028	0.030	0.030	0.040	0.040	0.044	0.062	0.040	0.048
$c_{pb} = 0.45$, $N = 140$	0.046	0.042	0.045	0.059	0.056	0.059	0.072	0.057	0.046
$c_{pb} = 0.45$, $N = 280$	0.055	0.049	0.054	0.066	0.062	0.062	0.077	0.067	0.061

Robustness up to the feasibility bound

Table 21 runs confounds inside the bounds. The G9 Type I error is 0.042 to 0.046, and its power at $c_{bm} = 0.30$ is 0.682 to 0.780.

Table 22 tests larger samples under prog-PB. At $c_{pb} = 0.45$ the G9 rejection rate rises from 0.044 at $N = 70$ to 0.059 at 140 and 0.062 at 280, and F1 is at 0.067.

A biomarker that predicts the rate of natural-history change

Table 23 shows the pathway that the feasibility bound does not limit. Under the null, G8 rejects at 0.068, G9 at 0.074 and F1 at 0.059, and the G9 mean estimate is +0.19 where the truth is zero.

Table 24 adds a patient-specific linear trend. G9 rejects at 0.043 under the null and 0.969 at $c_{bm} = 0.45$.

Table 25 adds random slopes to G9. The three variants differ from G9 by 0.010 or less in every cell.

Table 23: Stage 5, referee-variant runs: rejection rate (MCSE), mean interaction estimate on the calibrated scale and calibration ratio under a biomarker that predicts the rate of natural-history change (coefficient 0.3), Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 1,000 replicates (two-MCSE band at the null $[0.036, 0.064]$). The true value at the alternative is -3.6.

Cell	Method	Rejection	Mean estimate	κ
null ($c_{bm} = 0$)	G8	0.068 (0.008)	+0.21	0.96
	G9	0.074 (0.008)	+0.19	0.94
	F1	0.059 (0.008)	+0.11	0.98
$c_{bm} = 0.45$	G8	0.954 (0.007)	-3.42	0.96
	G9	0.954 (0.007)	-3.25	0.90
	F1	0.938 (0.008)	-3.00	0.97

Table 24: Stage 5, referee-variant runs: rejection rate (MCSE), mean interaction estimate and calibration ratio under a patient-specific linear trend in the natural-history component (SD 0.3 outcome units per week), Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 1,000 replicates.

Cell	Method	Rejection	Mean estimate	κ
null ($c_{bm} = 0$)	G8	0.042 (0.006)	-0.02	1.01
	G9	0.043 (0.006)	-0.01	0.99
	F1	0.040 (0.006)	-0.02	1.01
$c_{bm} = 0.45$	G8	0.970 (0.005)	-3.63	0.98
	G9	0.969 (0.005)	-3.45	0.92
	F1	0.955 (0.007)	-3.14	0.99

Table 25: Stage 5: rejection rate of G9 and its three random-effects variants on twelve cells (Hybrid, $t_{1/2} = 0.5$ week, $n_{sim} = 1{,}000$; $c_{bm} = 0$ rows are Type I error). The strength-0.6 rows were generated under the repaired matrix described above and are not evidence about the confound.

Cell	G9	+ slope on t	+ slope on D_e	+ both
none, $c_{bm} = 0.00$, $N = 35$	0.050	0.050	0.050	0.046
none, $c_{bm} = 0.00$, $N = 70$	0.043	0.044	0.043	0.041
none, $c_{bm} = 0.45$, $N = 35$	0.759	0.762	0.766	0.769
none, $c_{bm} = 0.45$, $N = 70$	0.969	0.970	0.969	0.970
prog-PB 0.6, $c_{bm} = 0.00$, $N = 35$	0.060	0.060	0.059	0.060
prog-PB 0.6, $c_{bm} = 0.00$, $N = 70$	0.047	0.046	0.047	0.047
prog-PB 0.6, $c_{bm} = 0.45$, $N = 35$	0.594	0.593	0.593	0.596
prog-PB 0.6, $c_{bm} = 0.45$, $N = 70$	0.860	0.859	0.858	0.858
prog-TV-const 0.6, $c_{bm} = 0.00$, $N = 35$	0.050	0.053	0.051	0.054
prog-TV-const 0.6, $c_{bm} = 0.00$, $N = 70$	0.044	0.044	0.043	0.040
prog-TV-const 0.6, $c_{bm} = 0.45$, $N = 35$	0.489	0.488	0.489	0.485
prog-TV-const 0.6, $c_{bm} = 0.45$, $N = 70$	0.772	0.772	0.773	0.768

Recommendation

For a Hybrid trial we recommend G9, the exposure-weighted predictor with an AIC-selected half-life, refitted by REML and tested with CR2. Its size is about 0.055 at $N \leq 70$, so the selection step costs about half a point of size. A fixed 1-week analysis half-life with CR2 holds nominal size for at most two points of power. The recommendation is conditional on the extended DGP and the parameters simulated. A biomarker that predicts the natural-history trend biases every analysis examined. A trial with such a biomarker needs a remedy that this study did not evaluate.